PLECTA: Overlap-aware instance segmentation of dense strand networks from binary masks

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Abstract

Many materials and tissues, including carbon-nanotube sheets, collagen and paper, comprise overlapping strands whose arrangement influences their effective properties. Measuring this geometry from micrographs requires individual strands to be traced through crossings and signal gaps, which merge distinct strands or split one strand into disconnected fragments. We present PLECTA (from Latin *plecta*, a braid), a deterministic method that segments individual strands using only a binary image whose foreground pixels mark strand centrelines. At each junction, PLECTA selects which strand segments continue into one another while allowing segments to terminate. It formulates this decision as an exact maximum-weight partial-matching problem, with an explicit penalty for unpaired segments and no restriction on the number of segments entering a junction. The output permits crossing pixels to belong to multiple strands. On synthetic scenes, PLECTA resolved strand identities more accurately than three released alternatives at every density tested. The same ranking held on an independently published generator run at its own settings. A separate stage uses the corresponding greyscale micrograph to estimate which strand lies on top at each crossing. The complete workflow is demonstrated on scanning electron micrographs of carbon-nanotube sheets, from raw images to individual strand instances with estimated crossing order, without manual annotations for training or as pipeline inputs.

Keywords: strand networks; instance segmentation; graph matching; geometric reconstruction; carbon nanotubes; scanning electron microscopy.

1 Introduction

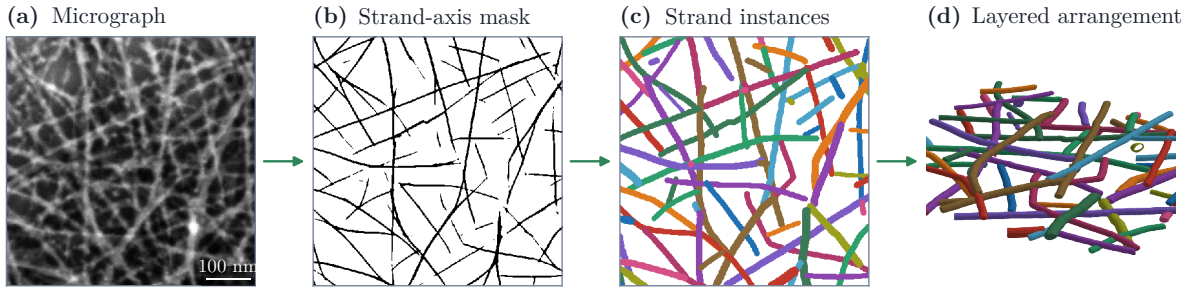


Figure 1. From micrograph to individual strands. **(a)** Input micrograph, a fixed central crop of one carbon-nanotube SEM field. **(b)** Upstream segmenter output. **(c)** PLECTA instance-segmentation output. **(d)** Depth-stage output.

Carbon-nanotube (CNT) sheets, collagen networks, paper and nonwoven textiles consist of overlapping strands whose arrangement influences their effective properties. Their stiffness,

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strength and conductivity depend on network geometry, constituent behaviour and inter-strand interactions. Load transfer between nanotubes, for example, can limit an assembly's stiffness and strength despite the high modulus of each tube [1–6]. Predictive models therefore require geometric descriptors of individual strands, whose identities must be recovered from the overlapping network visible in a micrograph.

Recovering those identities is an instance-segmentation problem in which the objects are long, open curves that overlap one another. Crossings merge distinct strands into one foreground component, whereas signal gaps split one strand into fragments whose tips may be indistinguishable from genuine ends. Segmentation must therefore determine which segments continue through each junction, which reconnect across a gap and which terminate.

We present a workflow that recovers individual strands and estimates their depth order from a single greyscale micrograph (Figure 1). First, a segmenter trained on synthetic image–mask pairs produces a binary mask marking the strand axes, without requiring manually annotated micrographs for training.

The central segmentation stage, PLECTA, then resolves individual strands from this mask alone. At each junction, it considers the possible strand continuations together while allowing genuine strand ends. It also reconnects fragments across signal gaps. This stage requires no training and is intended for networks whose strands approximately maintain their direction through crossings. Finally, a depth-ordering stage uses the segmented strands and the greyscale image to estimate which strand lies on top at each crossing. Identity is resolved in the image plane; depth ordering is a separate downstream stage, and no volumetric reconstruction is performed.

We evaluate PLECTA on synthetic scenes and annotated scanning electron microscopy (SEM) images of CNT sheets. It achieves higher strand-identity scores than the compared methods on the evaluated dense-network benchmarks.

Section 2 reviews related work, Section 3 describes the workflow and evaluation, Section 4 presents the results, and Section 5 discusses the conclusions and limitations.

2 Related work

Prior work relevant to the problem of Section 1 can be grouped into five categories: network-level measurement tools, greedy instance-segmentation methods, joint neighbourhood-optimisation methods, complete micrograph-to-measurement workflows, and monocular depth-ordering methods.

2.1 Network-level measurement tools

Existing methods address complementary aspects of the problem. Semantic segmentation, which labels every pixel with a class but does not distinguish one object from another, is mature: U-Net, named for its U-shaped encoder–decoder architecture, and its self-configuring descendants reliably separate strand foreground from background [7, 8], including in materials imaging [9, 10] and nanotube micrographs specifically [11–13]. Variants with explicit topology enhancement target the connectivity of that foreground [14], but a foreground mask still encodes no identity, so it inherits both failures rather than resolving them. Instance methods go further and label each object separately, but they assume those objects are compact, in materials microscopy [15] as in the general-purpose families: bounded blobs, centre-directed flows, star-convex outlines, overlapping ellipses, none of which describes a long open curve that may overlap another [16–20]. Ridge and vesselness filters, which score each pixel by how strongly the local image curvature resembles a line or a tube, sharpen curvilinear signal but assign nothing across a junction [21–23]. The established fibre-analysis packages return network statistics rather than resolved objects [24–28], as do image-level measures of coverage and alignment [29–31].

Volumetric imaging avoids the problem altogether, because strands that appear to meet in a projection lie at different depths and are therefore already apart. X-ray tomography of fibrous solids and of twisted ropes recovers individual fibres on that basis, by seeded watershed and morphological reasoning on the reconstructed volume [32, 33]. A single projection offers no such separation, because two crossing strands genuinely share the same pixels.

2.2 Greedy continuation from a mask

Recovering strands through crossings has been addressed repeatedly. Existing methods differ primarily in the evidence used for continuation and in whether earlier pairings can subsequently be revised.

A large class of methods follows a common procedure: skeletonise the mask, remove the junction pixels, and pair the resulting arms according to directional continuity. Each pairing is accepted as soon as it is scored. SIFNE, an extractor for filament networks in localization-microscopy images, reads its tip directions from the excised stubs and recovers dense microtubule networks from the mask alone [34]. Wegmayr et al. pair skeleton branches by lowest mutual angle [35]. GraFT (graph of filaments over time) covers a Frangi-filtered skeleton greedily, longest path first [36]. This is inexpensive, and where crossings are rare it suffices. Its weakness is structural: once an arm is paired, a later contradictory candidate cannot revise the pairing.

Two remedies have been tried. The first is to incorporate additional image evidence beyond the mask. The stronger of Wegmayr et al.'s two variants drops the angle rule and learns pixel embeddings on patches around each junction; it reads the greyscale image, and it must be trained on crops in which every fibre is already labelled individually. Even then it recovers tangled fibres at about half the precision of isolated ones, in data where two thirds of fibres cross nothing at all. Methods that pair ends with an orientation network, detect fibres with a region-based detector, or trace them recurrently [37–39] need the same labelled training data. Reading a mask and nothing else avoids that cost, a trade-off adopted in this paper. DNAi also uses only the mask but evaluates a larger set of candidate pairings, enumerating every pairing a junction allows from its endpoint angles; it does so exhaustively at degree three and four, and not at all above [40]. Liu et al. also read a mask alone, reconnect gaps and permit overlapping output, but pair termini greedily and depend on a trained orientation network [37].

2.3 Joint neighbourhood optimisation

The second is to resolve several candidate pairings together, rather than deciding each one alone. Basu et al. and DeFiNe both weigh a whole neighbourhood at once, which is what a dense node requires. Basu et al. delete every branch point from a spanning tree over centreline points, then reassemble the pieces with an integer program that penalises each join by its bending, iterating until the assignment converges; a join that a later fit contradicts can therefore still be undone [41]. DeFiNe covers an already-built weighted graph with the smoothest paths it can find, and lets two paths share an edge, so a crossing pixel can belong to both strands instead of being awarded to one [42]. Basu et al. require image intensities, and DeFiNe requires a graph produced by an upstream stage, rather than an axis mask. Both also reassemble at most three pieces at a time, which limits how much of a fragmented neighbourhood a single round can repair.

De et al. also weigh a whole neighbourhood at once, but reach a decision by propagating labels rather than reassembling pieces directly. Branches and junction contacts become a graph, and labels spread across it so that a decision at one node constrains its neighbours rather than resting on that node's own angles [43]. The labels, however, require initialisation. In their images they start at anatomical roots, the stained cell bodies of neurons or the optic disc of a retina, and the method returns as many strands as there are roots. A tangle of open-ended strands offers no such starting point, and how many strands it holds is precisely the unknown.

2.4 From micrograph to measurement

Complete workflows separate semantic segmentation from instance segmentation and validate the combined pipeline against a measurement task. Peters et al. join a U-Net-like segmentation trained on over a thousand annotated SEM micrographs to algorithms that trace fibres through intersections, and validate against human evaluators on fibre number, length and width [44].

Which strand lies on top at a crossing is a further question, and for nanostructured networks it is normally answered destructively, by focused-ion-beam sectioning and tomographic reconstruction [45], because a micrograph's intensities record surface topography rather than height. Monocular depth ordering has long derived pairwise depth from local occlusion evidence, assembled those relations into a depth-order graph and corrected the contradictions among them [46, 47], and single-view hair modelling resolves the layering of intersecting strands by analogous reasoning [48]. Amodal instance segmentation also predicts the hidden extent of each occluded object, guided in DBNet by estimated depth [49]. Those methods read the shape of the outlines where two objects overlap, which a bundle a few pixels across in a noisy micrograph does not reliably provide; the evidence used in this paper is instead the continuity of each strand's own brightness across the crossing.

Weighing a whole neighbourhood at once, with no seed and no strand count supplied in advance, is the problem addressed in this paper. Four of these methods are measured against PLECTA directly in Section 4.5, on identical masks and through one scorer.

3 Materials and methods

PLECTA turns a binary strand-axis mask into a set of overlap-aware strand instances, as shown in the workflow diagram in Figure 2. For validation and comparison against a known ground truth, the mask can be generated synthetically; masks derived from SEM images are also tested in this paper (Sections 3.1 and 3.2). PLECTA then builds a graph of arms, stubs and junction nodes and fits a tangent and curvature frame to each stub (Section 3.3.1). It scores candidate continuations (Section 3.3.2), solves exact partial matchings at junctions and across admissible gaps over eight frame-refitting rounds (Section 3.3.3), and converts the accepted links into overlapping strand instances (Section 3.3.4). Given a paired greyscale image, downstream stages estimate strand width and render ribbons (Section 3.4.1) and infer depth order at crossings (Section 3.4.2). Section 3.5 defines the measures all results are scored through and their statistical treatment.

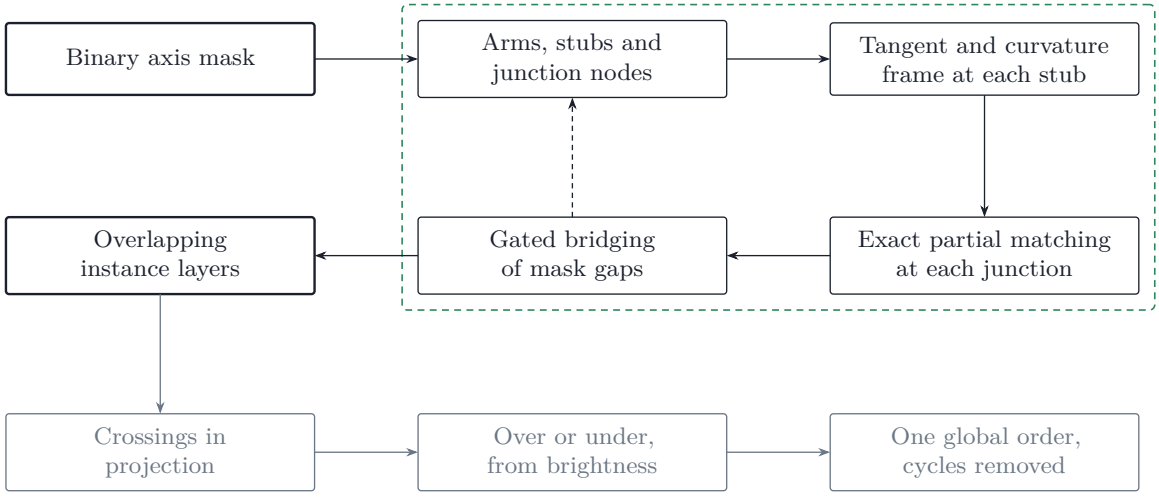


Figure 2. Scope of PLECTA. The dashed grouping stage maps a binary axis mask to overlapping instances; the loop denotes eight rounds of frame refitting and matching. The grey branch uses a paired greyscale image to infer depth order without changing strand identity.

3.1 Synthetic data

Strand centrelines are generated as random walks with Gaussian-smoothed angular increments, producing gradual bends with controllable curvature. The principal synthetic segmentation scenes are 512×512 pixels. Each full-width bundle is assigned a width sampled uniformly from 7 to 16 pixels, and strands are placed at random positions and orientations until their combined full-width area reaches the target areal coverage of 20, 30, 40, 50 or 60 % of the image area. Each scene provides strand centrelines $\mathcal{C}_k$, their corresponding pixel sets Ω_k, and four image representations: a clean one-pixel axis mask; a one-pixel axis mask degraded with gaps and raster irregularities; a full-width foreground mask; and an SEM-like greyscale image with strand-specific brightness, gradual intensity variation, blur and background noise. For depth evaluation (Section 3.4.2), a depth-resolved variant assigns strand heights and radii, prevents interpenetration and renders the scene from above. Occlusion, depth-dependent defocus and width broadening are included, with imaging cues randomised independently so that no single cue reliably determines crossing order.

3.2 CNT SEM data and mask sources

Three carbon-nanotube (CNT) sheet fields, imaged by scanning electron microscopy (SEM), were analysed, each comprising 1536×1024 pixels: B58_100, B58_110 and B58_300, containing 307, 430 and 325 annotated bundles, respectively. Each strand annotation represents a visible bundle, as individual nanotubes are unresolved at this magnification. The horizontal field width was 1.66 µm, corresponding to 1.08 $\mathrm{nm\,px^{-1}}$. Images were acquired using a Thermo Fisher Helios 5 UX DualBeam with an Elstar field-emission column in immersion mode and a through-lens detector, at 2.00 kV, 400 pA and a working distance of 4.3 mm. Each frame integrated 35 passes at a dwell time of 100 ns with drift correction.

The first author (O.A.) and an independent second annotator each annotated all 3 fields using an in-house tool that permits overlapping strand masks. Ridge snapping and spline smoothing assisted tracing, while strand connections and suggested gap bridges required manual decisions. Scores are reported separately against each annotator, with their agreement discussed in Section 5. Supplementary Section S7 describes the annotation procedure.

Two input masks were evaluated for each field: a skeletonised mask derived from the first author's reference and an automatically generated mask from the upstream CycleGAN (cycle-consistent generative adversarial network) and nnU-Net segmentation (Section 4.3) [8, 50]. These inputs assess segmentation from manually curated and automatically extracted strand axes, respectively. The automatic segmentation required no manual annotations for training or inference.

Strand segmentation was evaluated on the common centreline fragments visible within each input mask, using the metrics defined in Section 3.5.

3.3 PLECTA instance segmentation

Starting from a binary strand mask, PLECTA segments individual strands by resolving connections through junctions and across gaps. The following subsections describe the main steps: constructing the skeleton graph, scoring candidate connections and selecting compatible links, with iterative refinement as longer strands are assembled. For an input mask $M \in \{0,1\}^{H \times W}$, the output is a set of strand instances $\{I_1, \ldots, I_N\}$, with $I_n \subseteq F = \{x : M(x) = 1\}$; crossing pixels may belong to multiple instances. Segmentation uses the mask alone, while subsequent width and depth estimates also use the greyscale image (Section 3.4).

3.3.1 Skeleton graph and local geometry

PLECTA skeletonises the input mask using the Zhang–Suen thinning algorithm [51] and removes short spurs. Pixels where three or more skeleton branches meet are grouped into junction clusters, with nearby clusters merged into a single *node*. This accounts for crossings that span several pixels or lie too close together to resolve separately. The remaining strand segments form *arms*, each with a *stub* at either end.

Each stub is characterised by its tip position, outward tangent ($\mathbf{t}$) and signed curvature (κ), estimated from a short segment of the adjoining strand (Figure 3a). Subscripts identify the stub, so $\mathbf{t}_a$ and κ_a denote the tangent and curvature of stub a. Stubs with insufficient support for a reliable direction estimate are marked as unreliable. As arms are linked into longer chains, these estimates are updated using the additional strand geometry.

3.3.2 Pairing cost function

The cost C_{ab} of linking stubs a and b combines directional alignment, chord alignment and curvature continuity (Equation 1; Figure 3). The reversal angle θ, measured between $\mathbf{t}_a$ and $-\mathbf{t}_b$, is zero when the outward tangents form a straight continuation. The chord angles φ_a and φ_b measure how far each tangent turns towards the opposite tip, penalising lateral offset between otherwise aligned arms. The curvature mismatch $|\kappa_a + \kappa_b|$ penalises changes in bending; the sum reflects the opposite orientations of the outward frames. Gap links also incur a penalty for tip separation d:

$$C_{ab} = w_d\,\theta + w_t\,q(d)\,(\varphi_a + \varphi_b) + w_\ell\,\frac{d}{\ell_0} + 30\,w_\kappa\,|\kappa_a + \kappa_b|, \qquad q(d) = \min\Big(1, \tfrac{d}{d_0}\Big)\,, \tag{1}$$

where w_d, w_t, w_ℓ and w_κ weight the four terms, and ℓ_0 normalises the gap distance. Junction links set $w_\ell = 0$.

The reversal angle θ is independent of the tip-to-tip chord and remains useful when nearby tips make the chord direction sensitive to pixel quantisation. The factor $q(d)$ therefore reduces the chord-angle contribution at separations below d_0. The weights, distance scales ℓ_0 and d_0, and curvature scaling factor of 30 are fixed parameters listed in Supplementary Section S1. These costs quantify the suitability of individual connections; the next step selects a compatible set of links.

3.3.3 Candidate selection and iterative matching

Using the costs from Equation 1, PLECTA selects links jointly so that no stub is assigned to more than one connection. Leaving stub i unmatched incurs a penalty p_i. Connecting stubs i and j avoids both unmatched penalties but incurs the linking cost C_{ij} defined in Equation 1. The net benefit of the connection is therefore $p_i + p_j - C_{ij}$. A link is admissible only if $C_{ij} < p_i + p_j$. Exact maximum-weight partial matching then selects the set of disjoint stub pairs Γ with the greatest total benefit:

$$\max_{\Gamma} \sum_{(i,j)\in\Gamma} (p_i + p_j - C_{ij}). \tag{2}$$

Each stub can belong to at most one pair or remain unmatched, allowing genuine strand termination at junctions of any degree. The penalties therefore determine both candidate admissibility and the preference for linking. Because competing links are considered together, a locally favourable pair may be rejected when another combination gives a better overall assignment.

Junction and gap links use different penalties. Gap candidates must also satisfy distance and angular limits, since they connect strands across missing image evidence. Figure 4 illustrates candidate selection and matching at junctions and across gaps.

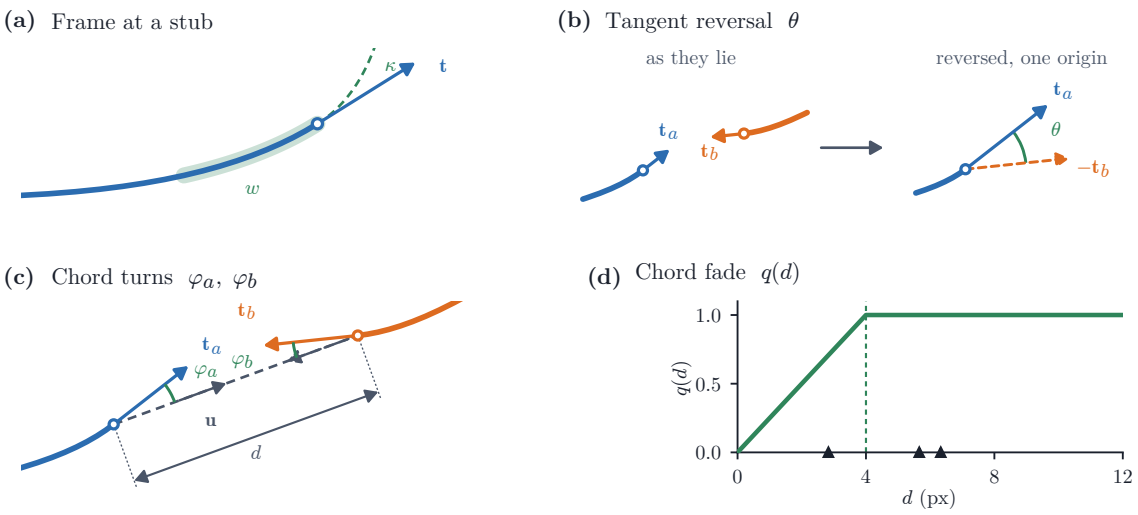


Figure 3. The stub frame and the terms of the pairing cost. **(a)** The least-squares arclength fit over window w, read outward from the tip at $s = 0$, giving outward tangent $\mathbf{t}$ and signed curvature κ; quadratic when the span allows, linear otherwise. **(b)** θ, in the two steps its definition takes: the outward tangents where they lie, then the same pair with $\mathbf{t}_b$ reversed and both read from one origin. A perfect continuation makes them antiparallel, so $\theta = 0$. **(c)** The chord turns φ_a, φ_b onto the chord $\mathbf{u}$ joining the tips. **(d)** The fade $q(d)$ against tip separation; triangles mark the six real separations at the crossing of Figure 4.

Matching proceeds iteratively, with tangent and curvature estimates updated along the assembled chains before each round. Early rounds use lower unmatched penalties and tighter gap-angle limits to restrict uncertain connections. These constraints gradually relax to their configured values, and the final round determines the output. Supplementary Section S1 lists the penalties, geometric limits and iteration schedule; Section 4.4 evaluates joint matching and the contributions of individual components.

3.3.4 Strand instance assembly

After the final matching round, connected components of the accepted arm links define individual strands. Each strand contains the pixels of its constituent arms and associated junction clusters, allowing intersecting strands to share crossing pixels. Accepted gap links connect fragments belonging to the same strand without adding pixels across the gap. These strand instances provide the input to the image-based stages described next.

3.4 Image-informed stages

The segmented strands are combined with the corresponding greyscale image to estimate their apparent widths and render ribbons (Section 3.4.1), and to infer crossing order (Section 3.4.2). These stages preserve the strand identities established during segmentation and do not affect the identity scores. They use fixed parameters (Supplementary Section S1) and are demonstrated here on SEM images, although the procedure accepts other greyscale images.

3.4.1 Width and ribbon rendering

Intensity profiles sampled perpendicular to each strand, away from junctions, provide estimates of apparent width and brightness, together with uncertainty and quality indicators. The centreline is then resampled and smoothed, with accepted gaps interpolated, to render a ribbon at the estimated width. Junction clusters are excluded from this rendering, and strands remain

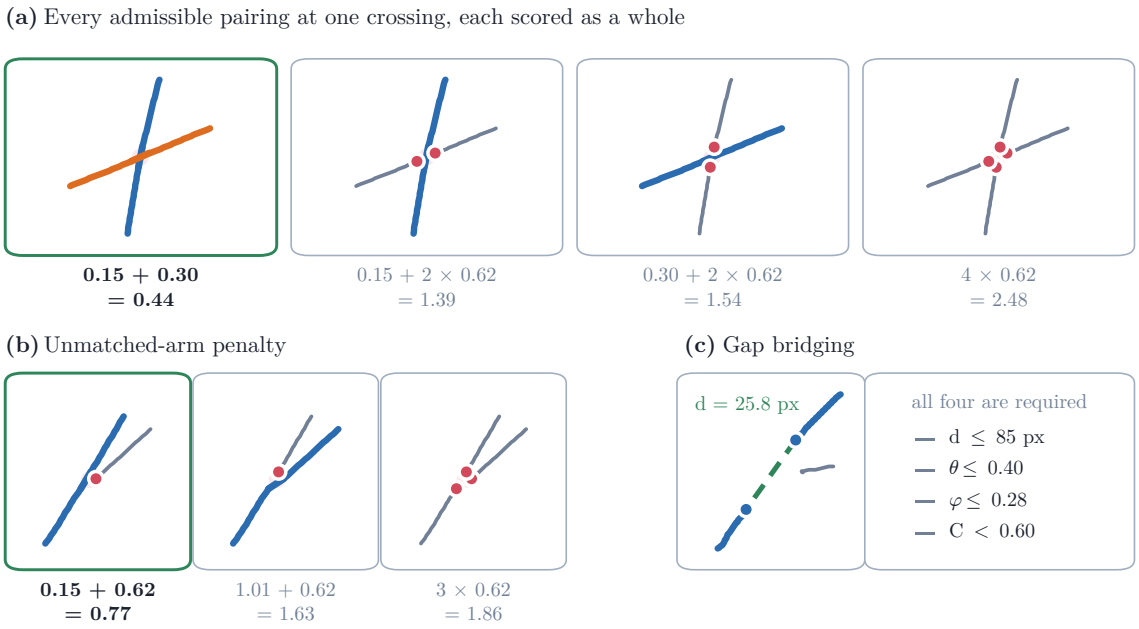


Figure 4. Candidate-link evaluation and matching outcomes, on the nodes where the decisions were made. Every arm is the skeleton the numbers were measured from, so every angle shown is the one the cost was computed on. **(a)** Every pairing the admissibility gate allows at a four-arm crossing, each scored as a whole: its edge costs plus the penalty $p_i = 0.62$ of each arm it leaves unmatched. The green frame marks the one returned; two further pairs exceed $p_i + p_j = 1.24$ and are excluded as candidates. **(b)** A three-arm node where an admissible edge is declined on the total rather than on its own cost. **(c)** A gap link, which must clear all four gates at once.

separate. The resulting ribbons describe their projected shapes; their relative positions in depth are estimated in Section 3.4.2.

3.4.2 Depth ordering and height assignment

Depth estimation proceeds in three steps: estimating local crossing orders, combining them into a consistent strand order, and assigning layers and heights.

1. Estimate crossing order. The method assumes that the upper strand retains its brightness through a crossing, while the lower strand is occluded. Intensity is sampled through the crossing (the *core*) and along each strand on either side (the *flanks*), excluding flank pixels occupied by the other strand (Figure 5a,b). The pooled core median m_c is compared with the flank medians m_i and m_j; the closer match indicates the upper strand.

The signed score s summarises this comparison (Figure 5c). It compares the two intensity mismatches, normalised by the larger of $|m_i - m_j|$ and twice the local noise estimate $\hat{\sigma}$. Its sign identifies the upper strand and its magnitude weights the evidence. Crossings with insufficient flank samples or a score below the threshold in magnitude remain undecided. Supplementary Section S2 gives the sampling procedure and score equation.

2. Combine crossing orders. Local estimates can conflict: A may lie above B, B above C, and C above A. These relations form a directed graph whose nodes are strands and whose edges carry weights $|s|$ (Figure 6). Within each connected component, a consistent strand order is sought with the objective of maximising the total weight of satisfied relations. Crossing orders are updated accordingly. Supplementary Section S2 describes the ordering procedure.

3. Assign layers and heights. The corrected relations form a graph without directed cycles. Its longest path determines the minimum number of layers needed to respect the crossing orders;

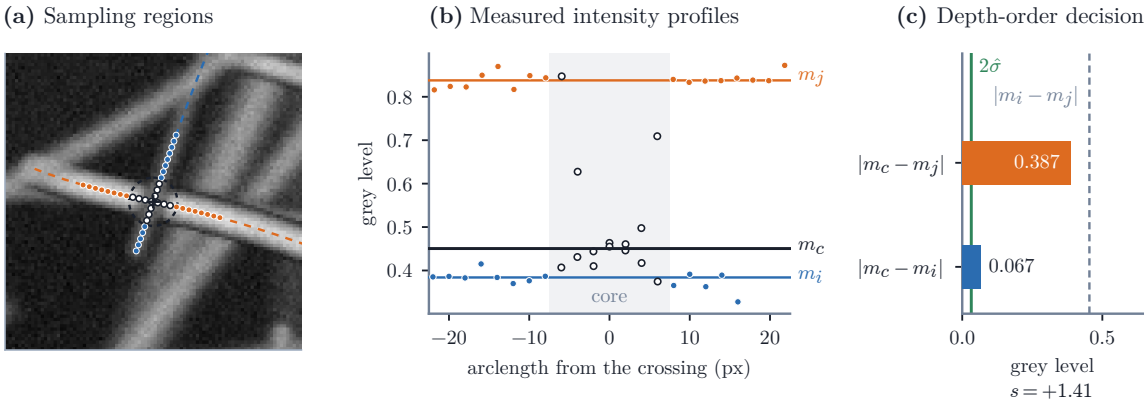


Figure 5. Local depth evidence at a crossing angle of 83.8°, selected for clarity. Positions and intensities are taken from the stage output. **(a)** Core samples (open dots) and flank samples (filled dots), with flanks kept clear of the other strand. **(b)** Intensity along each strand: the pooled core median m_c is closer to strand i's flank median than to strand j's, supporting i as the upper strand. **(c)** Intensity mismatches used to calculate s, normalised by the larger of the flank-median separation and twice the local noise estimate (Supplementary Section S2).

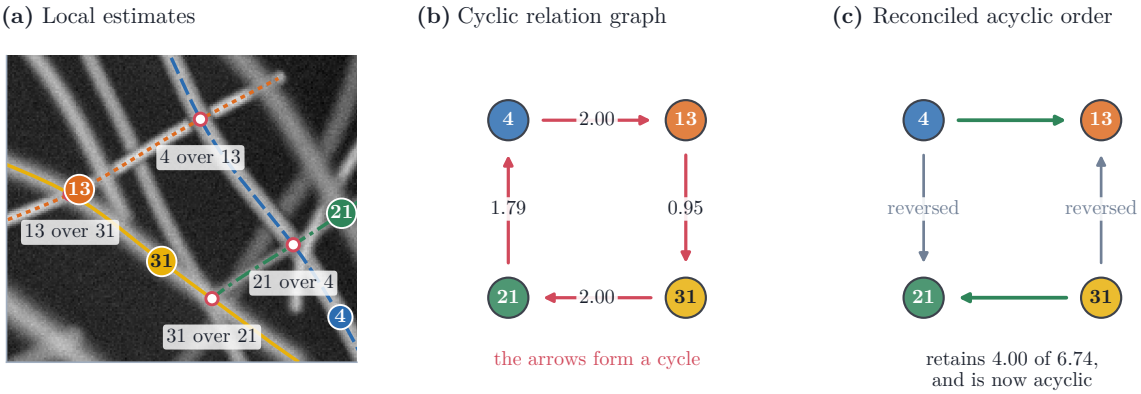


Figure 6. Resolving contradictory local depth estimates. **(a)** Four strands and their estimated crossing orders. **(b)** The corresponding graph, with arrows from the estimated upper strand to the lower strand. The directed cycle prevents a consistent height assignment. **(c)** The selected order retains the two highest-weight estimates and reverses the two lowest-weight estimates. Reversals are recorded with their original scores. The four strands shown belong to a seven-strand scene; layer assignments depend on the complete connected component.

strands without ordering constraints between them may share a layer. Under the compact-stack assumption, each strand is assigned the lowest height compatible with the chosen order and the estimated radii, preventing interpenetration.

Without a greyscale image, placement follows the predefined tie-break rather than measured crossing evidence. Section 4.2 evaluates depth ordering using the measures defined in Supplementary Section S5.

3.5 Metrics and statistical analysis

The primary segmentation metric is common-fragment pairwise F_1, which measures whether strand fragments are grouped consistently with the reference. Each input mask is divided into fragments between crossings, independently of the reference and predictions. The same assignment rule then gives each fragment a reference identity and a predicted identity. Pairwise precision penalises joining fragments from different reference strands, while recall penalises

separating fragments from the same strand. Figure 7 illustrates these cases and the conditions under which the score is undefined.

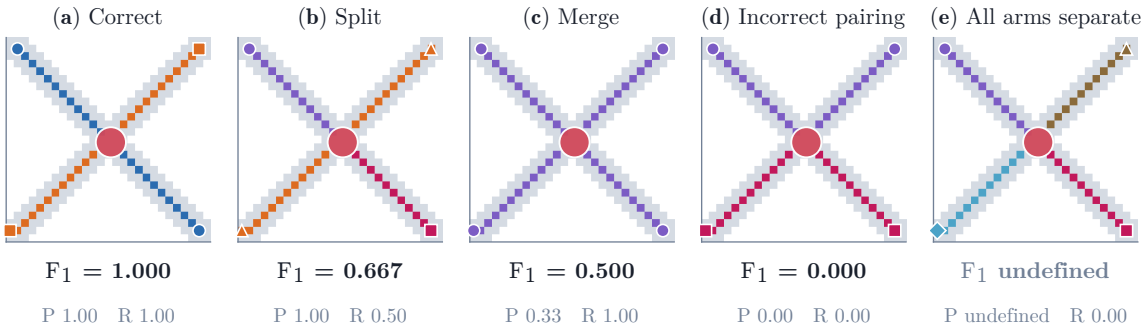


Figure 7. Pairwise scoring at a crossing. Removing junction pixels (red) leaves four arms and six fragment pairs. Tip markers identify arms assigned to the same strand. **(a)** Reference grouping. **(b)** A split reduces recall. **(c)** A merge reduces precision. **(d)** Incorrect pairing changes strand identities. **(e)** Separating every arm produces no predicted positive pair, so precision and F_1 are undefined.

Because junction pixels are excluded, this metric evaluates strand continuity through crossings rather than ownership of the crossing pixels themselves. Strands containing more fragments contribute more pairs and therefore carry greater weight. We also report the adjusted Rand index and variation of information [52, 53], together with detection F_1 and Crossing Fidelity, to characterise complementary aspects of segmentation. For the real SEM fields, Join F_1 evaluates joining decisions between nearby fragments, reducing the influence of differences in fragmentation between mask sources.

Synthetic segmentation means are accompanied by 95 % bootstrap confidence intervals, with scenes resampled within each density level. Comparisons between clean and degraded masks preserve scene pairing. For real SEM data, each image field is treated as one independent sample. Supplementary Section S3 provides the metric definitions, assignment rules and statistical procedures.

4 Results

We first evaluate strand segmentation on synthetic scenes (Section 4.1), then examine depth ordering (Section 4.2) and application to real SEM images (Section 4.3). Component ablations (Section 4.4) and comparisons with published methods (Section 4.5) assess the contributions and performance of the segmentation method.

4.1 Instance segmentation on synthetic images

These scenes carry exact reference identities, so every junction and gap decision can be checked against the grouping the generator recorded when it drew them. Across the 128 synthetic scenes, PLECTA achieved a mean common-fragment pairwise F_1 of 0.85 (95% bootstrap CI [0.84, 0.86]). Table 1 summarises segmentation scores overall and by areal coverage.

Performance decreased with scene density: mean F_1 fell from 0.91 [0.89, 0.93] at 20% coverage to 0.77 [0.76, 0.79] at 60%, monotonically through the intermediate strata. Per-scene runtime had median 1.12 s, ranging to 34.06 s on the densest scenes; Section 4.5 measures cost systematically.

PLECTA operates on the strand-axis mask, so segmentation difficulty depends on centreline density and geometry rather than areal coverage alone. The same areal coverage can arise from a different centreline density under another bundle-width distribution and may therefore produce different segmentation performance. Within the fixed 512 × 512-pixel frame and 7–16-pixel bundle-width range used here, areal coverage provides a consistent within-study stratification.

Table 1. Segmentation scores overall and by target areal coverage. VI denotes variation of information, for which lower is better; higher is better for the other metrics.

Group	n	F_1	Precision	Recall	ARI	VI split	VI merge	VI total
All	128	0.85	0.87	0.84	0.85	0.26	0.22	0.47
20%	32	0.91	0.93	0.90	0.91	0.14	0.09	0.23
30%	32	0.87	0.89	0.86	0.87	0.21	0.16	0.37
40%	32	0.86	0.86	0.85	0.85	0.24	0.22	0.47
60%	32	0.77	0.79	0.76	0.77	0.44	0.39	0.82

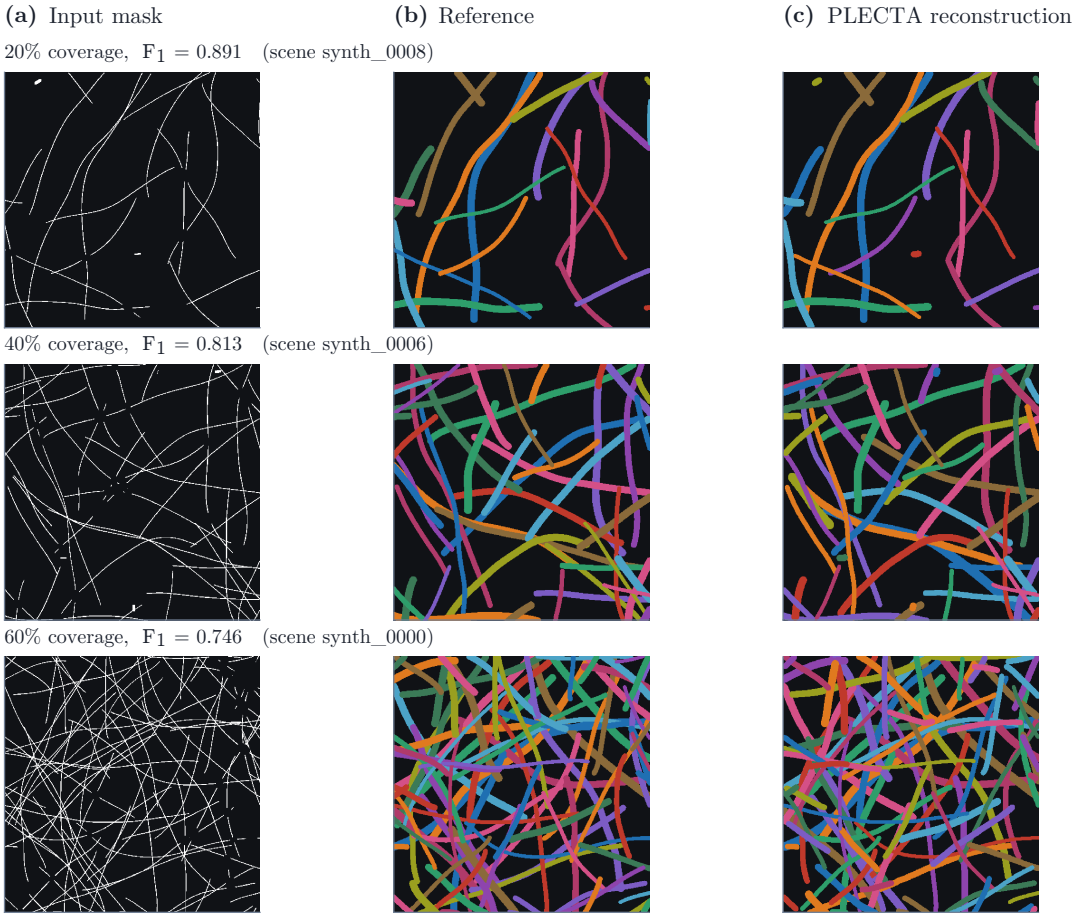


Figure 8. Segmentation at three coverage levels, each the scene closest to the median F_1 of its stratum, shown as a horizontal window of the scene. Columns: the degraded input mask, the overlap-aware reference, and PLECTA's strands at their fitted widths. The score beside each row is the grouping's, not the rendering's. Strand colours are arbitrary and not matched between columns.

To assess sensitivity to gaps and raster irregularities, we compared segmentation from clean and degraded masks of the same scenes. Across the 50 synthetic scenes, mean F_1 was 0.82 from degraded axes and 0.84 from the corresponding clean axes. The paired clean-minus-degraded change was 0.0224 (95% CI [0.0113, 0.0333]). Modelled gaps and raster irregularities therefore caused a small reduction in segmentation accuracy under this generator. The difference was greatest at low coverage and narrowed at 50–60 %, consistent with junction ambiguity becoming more influential at higher density.

Figure 8 shows representative segmentations at three coverage levels, selected by proximity to the median F_1 of each stratum.

4.2 Depth decomposition

The stages above segment strands in projection but do not establish which strand lies above at a crossing. After grouping is fixed, the depth stage uses the greyscale image to assess the continuity of each strand's brightness across the crossing relative to local noise. It reconciles these local estimates into a globally consistent vertical order and assigns non-interpenetrating heights.

The stage was evaluated on 12 depth-resolved scenes from Section 3.1, at two areal coverages and under two input conditions. Reference 2-D geometry isolates the depth decision, whereas binary axis masks include segmentation error. Table 2 reports both using the measures defined in Supplementary Section S5. Its crossing rows score local over/under decisions before reconciliation, once for every crossing; its stacking rows score the final order, once for every pair of strands.

Table 2. The depth stage on 12 synthetic scenes whose true vertical order is known. Chance is 0.5 for every accuracy shown.

	Given 2-D geometry		Segmented 2-D	
Areal coverage	20 %	40 %	20 %	40 %
At each crossing				
Projected crossings per scene	28	150	28	150
Crossings recovered	1.00	1.00	0.86	0.75
Order accuracy, all crossings	0.81	0.74	0.73	0.56
restricted to crossings the stage decided	0.89	0.88	0.96	0.91
Declined, of crossings the stage found	0.08	0.16	0.12	0.18
In the resulting stack				
Depth order, all strand pairs	0.67	0.71	0.70	0.72
restricted to pairs that cross	0.86	0.86	0.91	0.85
Pairs within one crossing component	0.39	0.84	0.36	0.75
Layers in the reference	5.2	8.7	5.2	8.7
Layers recovered	5.2	8.3	5.3	8.0
Layer agreement	0.66	0.42	0.70	0.38

Among crossings for which an order was assigned, accuracy ranged from 0.88 to 0.96 across both input conditions and densities. The all-crossing scores were lower (0.56–0.81) because they include crossings the 2-D stage did not recover and those for which the depth stage abstained. The table reports recovery and abstention rates alongside these scores.

Global order is limited to evidence that can travel through the crossing graph. Only 0.36 to 0.84 of strand pairs lie within the same connected component, so the relative order of the remaining pairs is not identifiable from the available crossing evidence. All-pair order accuracy was consequently 0.67–0.72, compared with 0.85–0.91 for pairs that cross. Layer agreement likewise fell from 0.70 to 0.38 as coverage doubled, although the recovered number of layers remained close to the reference.

Figure 9 shows what the stage produces on one synthetic scene, from the binary axis mask through to the film the reconstruction implies; Figure 10 is the same sequence on a real CNT SEM field, which no depth ground truth can score.

4.3 Instance segmentation on CNT SEM images

An upstream segmenter produces the binary strand-axis mask from each SEM micrograph (Figure 11), and PLECTA uses that mask alone.

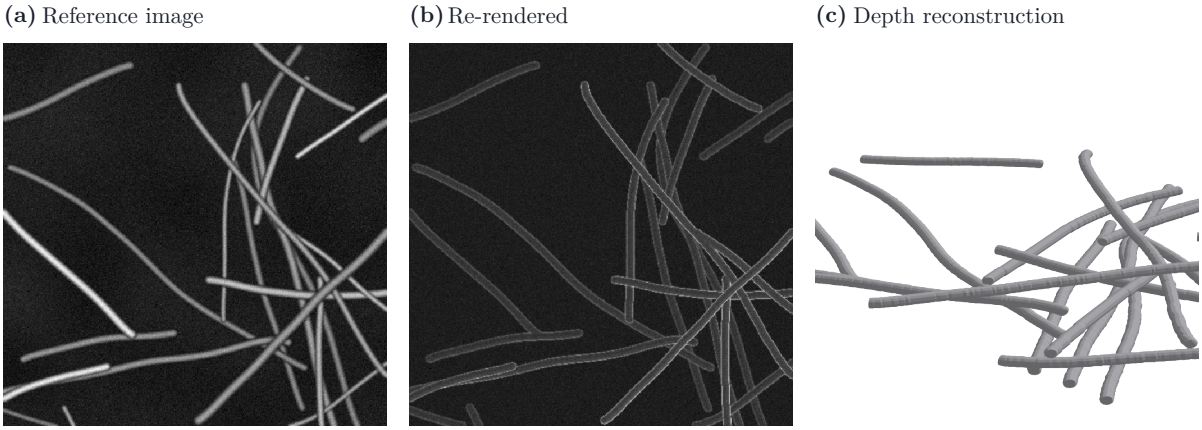


Figure 9. Depth reconstruction of a synthetic scene from its binary axis mask. The panels show the rendered greyscale image, a forward-model re-rendering from the reconstructed depths and radii, and the inferred strand stack. The re-rendering is a visual consistency check, not a scored quantity. The scene is 512 px across in the generator's pixel units.

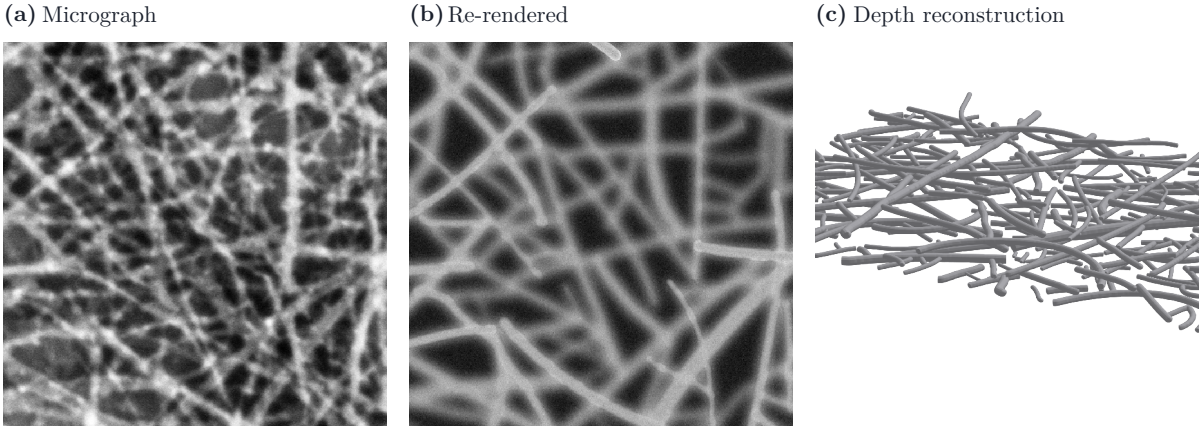


Figure 10. The same comparison on a real CNT SEM field, reconstructed from the reference axis mask. The crop is 512 px across, about 0.55 µm of field B58_100. Real SEM provides no depth ground truth, so the depth order is shown qualitatively; the forward-model appearance parameters are fitted to this micrograph.

We evaluated three manually annotated CNT-sheet fields under two input conditions: the skeleton of the reference annotation and the automatic nnU-Net axis [50]. Both were scored against the reference annotation, so the conditions differ only in the input mask. The reference-derived axis tests grouping from curated real-image geometry; the automatic axis evaluates the complete image-to-instances workflow. Figure 12 shows both conditions on one whole field.

Table 3 reports each field and condition. Mean F_1 was 0.89 from the reference-derived axis and 0.46 from the automatic axis. The reference-derived result establishes grouping on axes from real micrographs; the automatic result measures sensitivity to the input mask. With only 3 fields, these results are descriptive and the field is the independent sample.

The automatic axes were thicker, displaced and more fragmented than the reference-derived axes (Figure 12), and the additional errors were mainly missed links rather than erroneous mergers. Join F_1 narrows but does not remove the difference between the conditions. On B58_110, for example, the automatic axis gave join precision 0.68 and recall 0.53.

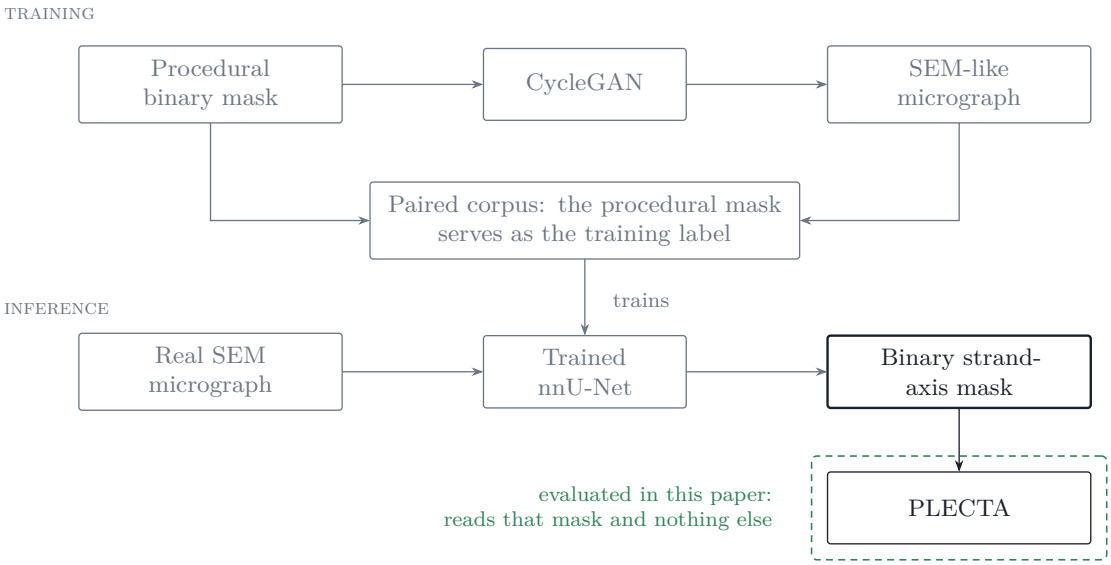


Figure 11. Upstream generation of the binary strand-axis mask. The segmenter is trained on procedurally generated image–mask pairs and requires no manual annotations for training or inference. Only the PLECTA stages inside the dashed boundary are evaluated here; the segmenter has its own evaluation [50].

Table 3. Segmentation from reference-derived and automatic axes in three CNT SEM fields. Coverage is the fraction of the frame covered by full-width reference strands. Metrics are defined in Section 3.5; detection F_1 uses IoU 0.1 and tolerance 2 px.

Field	Coverage	Input axis	F_1	Join F_1	Det. F_1
B58_100	44.0 %	Manual-derived	0.89	0.93	0.94
		U-Net	0.46	0.56	0.63
B58_110	46.7 %	Manual-derived	0.84	0.91	0.93
		U-Net	0.45	0.59	0.60
B58_300	41.5 %	Manual-derived	0.95	0.96	0.96
		U-Net	0.47	0.62	0.58

These three fields cannot disentangle segmentation error from ambiguity in manual strand identity. When PLECTA was given the independent annotation's skeleton instead of the automatic axis, it achieved F_1=0.38 against the reference. This lies within the direct inter-annotator agreement range of 0.36–0.53, showing that plausible manual readings can differ substantially where local occlusions obscure strand continuity. The automatic-axis result should therefore be interpreted relative to this reference ambiguity, as discussed in Section 5.

4.4 Component ablations

The real-SEM results show the effect of input-mask quality. To isolate the contribution of PLECTA's segmentation components, we removed one component at a time on the development scenes. Table 4 also includes two controls that replace, rather than remove, part of the method.

The largest losses followed removal of the junction chord-turn term, gap bridging and junction-cluster consolidation. The remaining rows test the curvature term, graph construction and iterative refinement; their smaller effects indicate that these components refine the core segmentation rather than replace it.

The first control replaces joint junction matching with greedy arm pairing while retaining the rest of PLECTA, showing the cost of decisions that are locally cheap but jointly incompatible. The second replaces the linker with a fixed-gate greedy endpoint-pairing baseline.

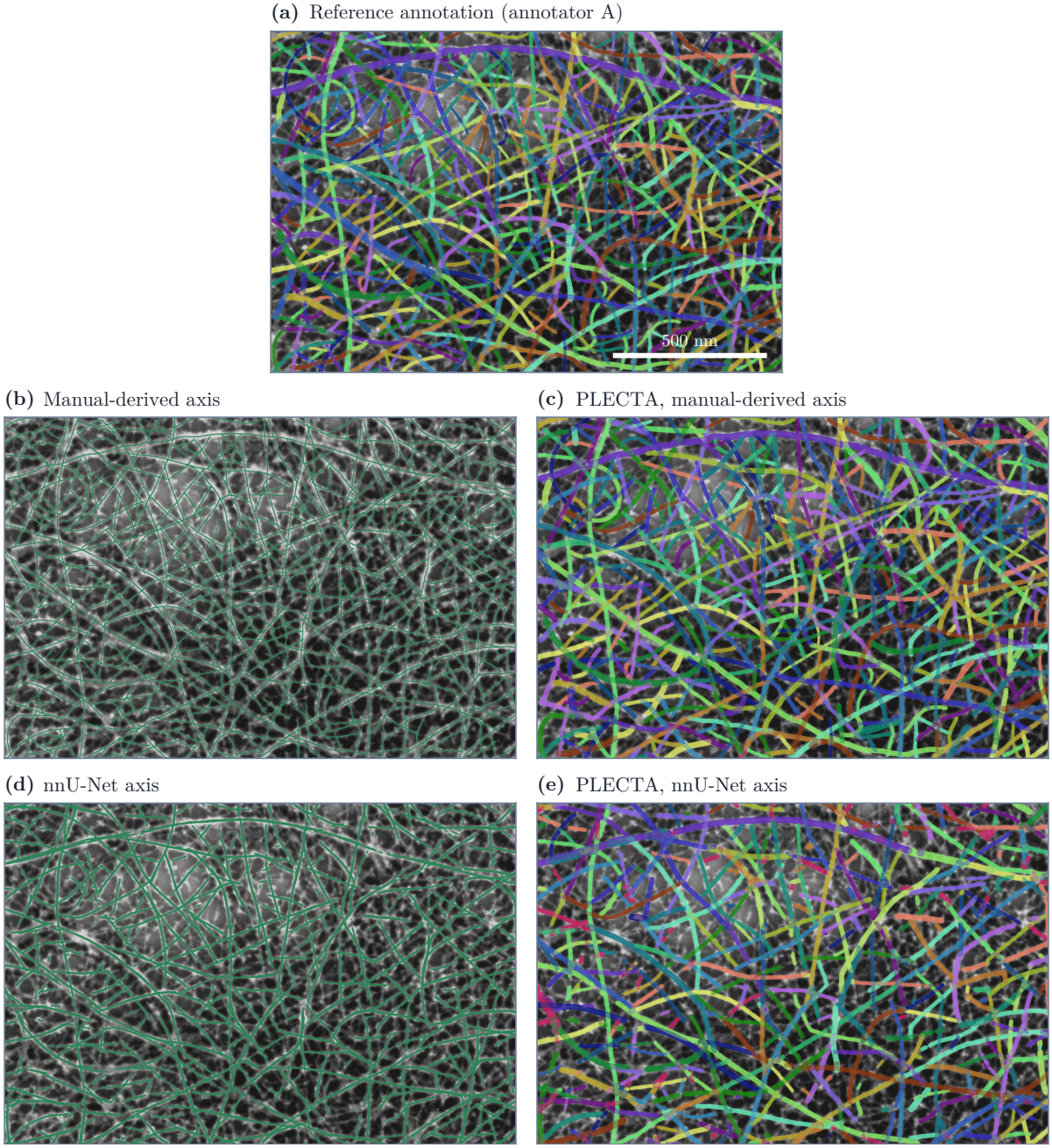


Figure 12. Segmentation of the whole B58_110 field from two input-mask sources; the scale bar in (a) is 500 nm. B58_110 is the densest of the three fields and the lowest-scoring on the reference-derived axis. **(a)** Reference annotation; **(b)** reference-derived axis; **(c)** PLECTA from that axis; **(d)** automatic axis; and **(e)** PLECTA from that axis with the same parameters. Input axes are dilated by one pixel for legibility. Predicted strands take the colour of their matched reference strand; crimson indicates no match.

4.5 Comparison with published methods

The ablations identify the components that support segmentation. We next compare the complete method with four published approaches on the 50 synthetic scenes. The two methods that accept binary axes were also evaluated on the three annotated CNT SEM fields of Section 4.3. All methods receive the same binary masks and are scored by the same pipeline. The comparison therefore concerns overlap-aware grouping of dense, crossing strand axes, not performance

Table 4. Component ablations and replacement controls on development scenes. Each ΔF_1 is relative to PLECTA evaluated on the corresponding scene set.

Variant	F_1	ΔF_1	ARI
Fixed configuration	0.87	+0.000	0.86
No joint junction solve	0.78	-0.080	0.77
No gap bridging	0.73	-0.139	0.72
Single round	0.85	-0.020	0.84
No chain-extended frames	0.85	-0.021	0.84
No round schedule	0.86	-0.004	0.86
No curvature term	0.86	-0.005	0.86
No short-connector node merging	0.85	-0.022	0.84
No junction-cluster consolidation	0.80	-0.063	0.80
No junction chord-turn term	0.68	-0.186	0.67
Whole linker replaced	0.68	-0.145	0.67

in each method's native application. DNAi [40], GraFT [36] and SIFNE [34] use released implementations; Basu et al. [41] is reimplemented because no implementation was available.

Table 5. PLECTA and four published methods on identical masks, ordered by F_1. The upper block uses degraded synthetic axes; the lower uses reference-derived axes from the CNT SEM fields of Section 4.3. Variation of information is in bits and lower is better; higher is better for the remaining metrics.
* our reimplementation; no implementation of that paper was found. Supplementary Section S4 details it.

						Crossing	Median s/scene	
Method	F_1	Precision	Recall	ARI	VI total ↓	fidelity	at 20 %	at 60 %
Synthetic scenes, degraded axis								
PLECTA	0.82	0.83	0.81	0.82	0.57	0.81	0.14	4.09
SIFNE	0.71	0.77	0.67	0.70	0.93	0.77	3.42	5.68
Basu*	0.54	0.66	0.47	0.53	1.37	0.76	2.56	54.20
GraFT	0.49	0.67	0.40	0.48	1.56	0.68	2.44	29.74
DNAi	0.34	0.46	0.28	0.33	2.11	0.58	0.04	0.56
Real CNT SEM fields, manual-derived axis								
PLECTA	0.89	0.86	0.93	0.89	0.33	0.93		
SIFNE	0.87	0.86	0.88	0.86	0.37	0.92		
Basu*	0.76	0.74	0.78	0.76	0.59	0.89		

For each released method, the mask was supplied at the start of its identity-assignment stage so that segmentation did not affect the comparison. All methods therefore received the same masks and were scored by the same code. Configurations were selected on the same development scenes used for PLECTA, separate from the scored ones (Supplementary Section S6), rather than left at their defaults. The ranking also holds on GraFT's independently published generator, reported below. The injection points and configurations are documented in Supplementary Section S4.

Under these comparator-favouring settings, PLECTA ranked first at every density and under both mask conditions (Figure 13); SIFNE was the strongest comparator. DNAi was designed for sparse DNA-fibre micrographs and recovered 0.66 from clean axes at 20 % coverage. Basu ranked second on Crossing Fidelity but substantially lower on pairwise F_1, indicating that curvature-based junction matching alone is insufficient for the full segmentation task. Detailed results are given in Supplementary Section S4.

Other regimes. On GraFT's released generator, run at its published settings over 40 scenes, PLECTA again achieved the highest F_1 at every density (0.92 overall, compared with 0.48 for

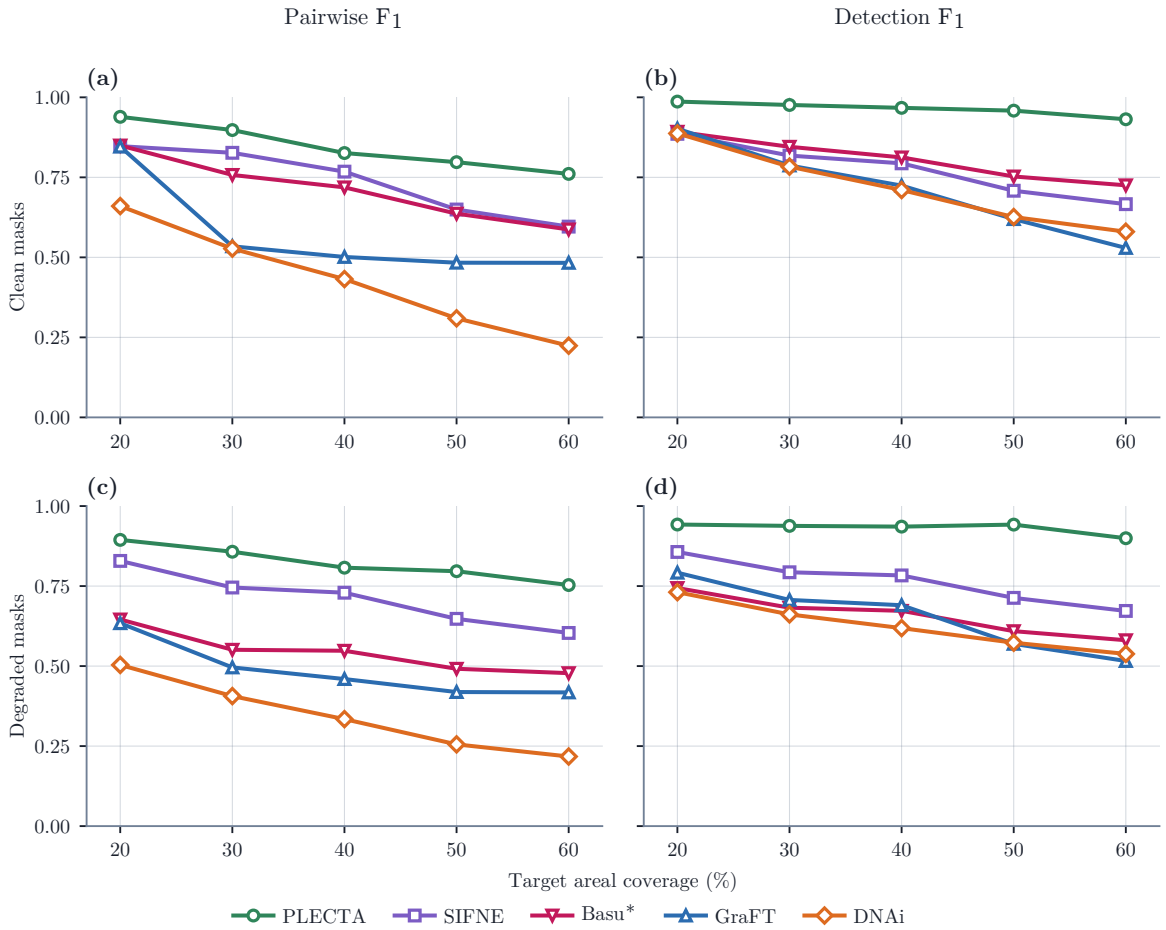


Figure 13. Five methods on identical masks, by coverage. The upper row uses clean axes and the lower row degraded axes. Columns: pairwise F_1, which counts pairs of fragments, and detection F_1 at IoU 0.1 with a 2 px tolerance, which counts objects (Section 3.5). The two disagree where a method fragments, since only the first is quadratic in fragmentation. Crossing Fidelity is reported in Table 5. * our reimplementation; no implementation was available.

GraFT and 0.35 for DNAi). SIFNE released no generator; on a recreation of its sparse cobweb topology, PLECTA and SIFNE were similar at low density and separated only slightly as density increased. This supports crossing entanglement, rather than strand content alone, as the relevant difference, although the recreated topology is weaker evidence than a released generator.

Cost. Table 5 reports warm, single-threaded per-scene times. Because the implementations are independently optimised, these times reflect implementation as well as algorithm.

Real SEM. On reference-derived axes from the three CNT fields, PLECTA achieved F_1=0.89, compared with 0.87 for SIFNE and 0.76 for Basu, and ranked highest on every reported measure. The smaller margin than on synthetic scenes is consistent with these curated axes containing fewer gaps, fragmentation and placement errors. Together, the ablations and comparisons show PLECTA's advantage for matched-mask segmentation in dense crossing networks; the scope of that claim is considered in the conclusions.

5 Conclusions

Micrographs of strand networks show crossings that merge distinct strands and gaps that split one strand into fragments, so each strand's identity is lost in the image. Recovering it is what makes per-strand length, orientation and curvature measurable. PLECTA recovers it from

a binary image marking strand centrelines alone, deciding at each junction which segments continue into one another and allowing a strand to end rather than be forced onward. Crossing pixels are assigned to every strand passing through them. PLECTA's grouping core requires no training data and produces identical output from identical input masks. The upstream segmenter is trained on synthetic image–mask pairs, so the full workflow requires no manually annotated training micrographs.

- Across 128 synthetic scenes, PLECTA achieved a mean pairwise F_1 (agreement on whether two fragments belong to the same strand) of 0.85 [0.84, 0.86], falling from 0.91 to 0.77 as areal coverage, the fraction of the frame the strands fill, rises from 20 to 60 %.
- With every method given the same input mask, PLECTA achieved the highest scores among the evaluated methods across the tested densities. Its lead was also retained on GraFT's released generator at published settings.
- The ablation analysis identified the junction term penalising turning, and the reconnection of fragments across signal gaps, as the largest contributors to segmentation performance.
- The greyscale depth stage ordered the crossings it did not decline with 0.88–0.96 accuracy, against chance level 0.5. Its global order is limited to strands connected through crossings.

Scope and limitations.

- PLECTA assumes directional continuity through crossings and remains sensitive to upstream mask errors. It can bridge short gaps but cannot recover omitted strands or reliably correct false axes.
- The real-material evaluation is a case study of three CNT SEM fields. The automatic-axis mean F_1 of 0.46 lies within the range over which two human annotators agree with each other, 0.36–0.53, so it should be interpreted alongside ambiguity in defining strand identity through local occlusions.
- The comparisons test all methods on the same binary masks in the dense, crossing networks studied here. They show PLECTA's advantage in this setting, but do not assess each method in the imaging application for which it was originally developed.

Outlook.

- A more accurate upstream segmenter, trained on synthetic data better matched to real micrographs, would improve the masks PLECTA receives.
- CNT sheets stacked in a known order, or focused-ion-beam cross-sections of the imaged region, would give the true crossing order for the depth stage; a real development set, kept separate from the evaluation fields, would let parameters be tuned on imaging rather than on synthetic scenes alone.
- The workflow remains to be tested on other materials, magnifications and SEM applications.
- The segmented strands, with their depth order and widths, can serve as input geometry for mesoscale mechanical simulations.

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Data and code availability

The PLECTA source, complete fixed configuration, compact seed manifest, and aggregate machine-readable evidence are available at `https://github.com/Jorallan/PLECTA`; the version used in this paper is archived at Zenodo [54]. A graphical application that performs the same segmentation on a supplied mask is available at `https://github.com/Jorallan/PLECTA_APP`. The annotation tool of Section 3.2 is at `https://github.com/Jorallan/ImageAnnotator`; it is listed because the machine-assisted annotation protocol described there is implemented in it. The manual annotations of the three SEM fields, from both annotators and in the form the scorer reads, will be deposited in 4TU.ResearchData. Generated scenes and detailed per-scene records are available from the corresponding author on reasonable request.

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